

Saturation Temperature and Pressure Lab

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Synopsis

The relationship between the saturation temperature and absolute pressure of water was investigated using an Armfield TH3 saturation-pressure apparatus. Nine equilibrium-stage readings were recorded from 50.00 kPa to 450.00 kPa gauge pressure. The measured atmospheric pressure of 101.65 kPa was added to each gauge reading. The original PT100 resistance readings were processed in two stages using the Armfield manual: the console indication was corrected with Data Sheet 1, then converted to temperature with Data Sheet 2. Linear interpolation was used between adjacent tabulated values.

Both the experimental and published steam-table data followed a smooth power relationship of the form $T = AP^n$. The experimental fit was $T = 31.218P^{0.2527}$, compared with $T = 31.019P^{0.2557}$ for the direct steam-table values. Experimental temperatures were below point-specific interpolated theoretical values at all nine pressures. The mean signed error was -1.03% , and the largest absolute error was 1.89% . The close exponents and small errors indicate that the experiment reproduced both the form and magnitude of the saturation relationship. The remaining differences are consistent with the assumptions of complete air purging, thermal equilibrium and valid linear interpolation, together with heat loss, probe lag and instrument resolution. Repeated readings at each pressure would be required to separate repeatability from systematic bias.

1 Aim

To measure the relationship between the saturation temperature and absolute pressure of water, determine an empirical power-law model, and assess the accuracy of the apparatus by comparison with published saturated-water data.

2 Theory and Data Processing

At liquid–vapour equilibrium, saturated liquid water and saturated steam have one saturation temperature for a given absolute pressure. Increasing pressure raises this temperature and produces a smooth, nonlinear saturation curve. The electronic pressure reading was gauge pressure, therefore

$$P_{\text{abs}} = P_1 + P_{\text{atm}}, \quad P_{\text{atm}} = 101.65 \text{ kPa}. \quad (1)$$

The PT100 processing was a correction and reference-table conversion, not a new two-point calibration. Armfield Data Sheet 1 corrects the indicated resistance for imbalance in the console resistance bridge. For an indicated resistance R_m between adjacent entries $R_{m,a}$ and $R_{m,b}$,

$$R_c = R_{c,a} + \frac{R_m - R_{m,a}}{R_{m,b} - R_{m,a}}(R_{c,b} - R_{c,a}). \quad (2)$$

Data Sheet 2 then relates corrected resistance to PT100 temperature. Linear interpolation between adjacent corrected-resistance values was used:

$$T_{\text{exp}} = T_a + \frac{R_c - R_{c,a}}{R_{c,b} - R_{c,a}}(T_b - T_a). \quad (3)$$

Data Sheet 1 explicitly permits interpolation. Data Sheet 2 instructs the user to select the closest entry; interpolation across its 2 °C increments was adopted here as a stated processing assumption to retain the resolution of the recorded resistances and avoid rounding every result to the nearest 2 °C.

Over the measured range, both datasets were represented by

$$T_{\text{sat}} = AP_{\text{sat}}^n, \quad (4)$$

where A and n were obtained by a log-space least-squares fit. Direct values from saturated-water pressure Table A-5 were used for the comparison curve. For the error calculation only, the theoretical temperature at each experimental pressure was found by linear interpolation:

$$T_{\text{th}} = T_a + \frac{P_{\text{abs}} - P_a}{P_b - P_a}(T_b - T_a). \quad (5)$$

The signed percentage error was

$$\text{Error (\%)} = \frac{T_{\text{exp}} - T_{\text{th}}}{T_{\text{th}}} \times 100. \quad (6)$$

3 Equipment

The equipment comprised an Armfield TH3 saturation-pressure apparatus with a 2.40 L boiler, sight glass, closed pipe loop, cartridge heaters, control console, filling and pressure-relief valves, a PT100 platinum resistance thermometer, electronic and Bourdon pressure gauges, a barometer, and de-ionised water. The manual gives a normal operating range of 0–7.00 bar gauge and a maximum allowable pressure of 8.00 bar gauge.



Figure 1: Armfield TH3 boiler and pipe loop used in the 2026 experiment. The photograph shows the boiler, pressure gauge, viewing port, valves and sensor wiring of the apparatus used.

4 Method

1. With the drain and calorimeter isolating valves closed and console power off, the boiler was filled with de-ionised water to halfway up the sight glass.
2. The water was heated to vigorous boiling with the filling point open. Steam was vented until the temperature indication stabilised, allowing air to be expelled from the loop.
3. The filling valve was closed. The apparatus was then treated as a sealed, approximately constant-volume system containing a saturated water–steam mixture. Heater input was raised for approximately two minutes and reduced.
4. Gauge pressure and PT100 resistance were recorded after the indication stabilised at nine stages from 50.00 kPa to 450.00 kPa gauge pressure.
5. Gauge readings were converted to absolute pressure. Each indicated resistance was corrected using Armfield Data Sheet 1 and converted to temperature using Data Sheet 2.
6. Experimental points and direct Table A–5 values were plotted on the same axes and fitted separately. Theoretical temperatures were then interpolated at the nine experimental pressures for the point-by-point error calculation.
7. After the final reading, the heaters were switched off and the isolating valve was opened before cooling to prevent a damaging partial vacuum.

5 Results

5.1 Worked calculation

For the first reading, $P_1 = 50.00 \text{ kPa}$ and $R_m = 142.30 \Omega$:

$$P_{\text{abs}} = 50.00 + 101.65 = 151.65 \text{ kPa}, \quad (7)$$

$$R_c = 142.32 + \frac{142.3 - 142}{143 - 142}(143.54 - 142.32) = 142.69 \Omega. \quad (8)$$

Data Sheet 2 gives 142.29Ω at 110°C and 143.04Ω at 112°C , therefore

$$T_{\text{exp}} = 110 + \frac{142.686 - 142.29}{143.04 - 142.29}(112 - 110) = 111.056^\circ\text{C}, \quad (9)$$

$$T_{\text{th}} = 111.35 + \frac{151.65 - 150}{175 - 150}(116.04 - 111.35) = 111.660^\circ\text{C}, \quad (10)$$

$$\text{Error} = \frac{111.056 - 111.660}{111.660} \times 100 = -0.541\%. \quad (11)$$

5.2 Processed results

Table 1: Original measurements, Armfield PT100 conversion, theoretical temperatures and errors.

P_1 (kPa)	P_{abs} (kPa)	R_m (Ω)	R_c (Ω)	T_{exp} ($^{\circ}\text{C}$)	T_{th} ($^{\circ}\text{C}$)	Error (%)
50	151.65	142.3	142.686	111.06	111.66	−0.54
100	201.65	144.8	145.788	119.27	120.46	−0.98
150	251.65	146.9	148.452	126.35	127.62	−0.99
200	301.65	148.7	150.780	132.55	133.70	−0.86
250	351.65	150.1	152.633	137.47	139.02	−1.11
300	401.65	151.1	153.964	141.04	143.75	−1.89
350	451.65	152.6	155.986	146.44	148.03	−1.07
400	501.65	153.8	157.634	150.86	151.95	−0.71
450	551.65	154.6	158.744	153.82	155.57	−1.12

5.3 Saturation curves and fitted equations

Figure 2 compares the processed experimental data with the direct pressure–temperature pairs from Table A–5. Interpolated theoretical points were not used to build either trendline.

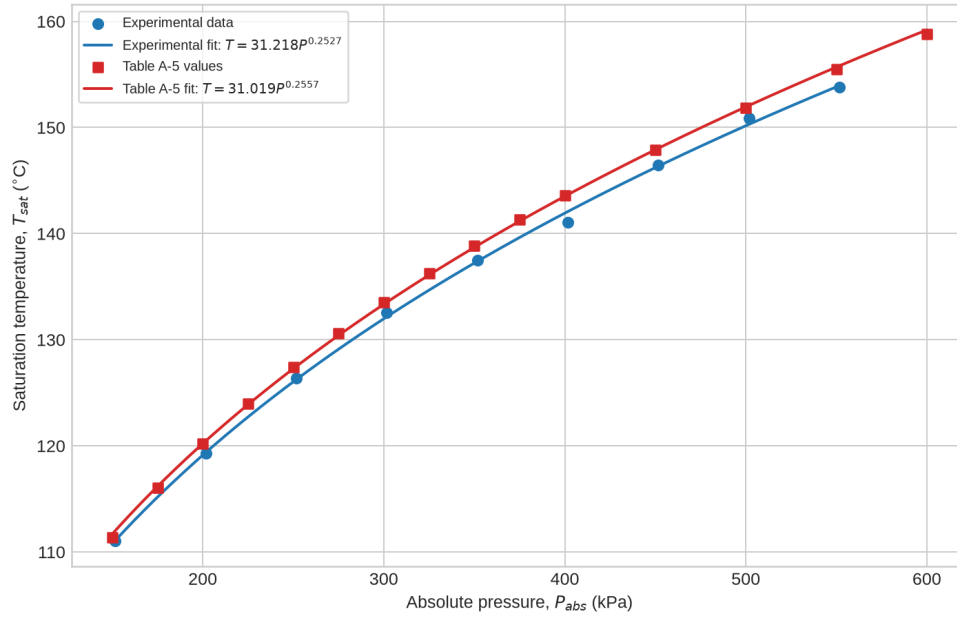


Figure 2: Experimental temperatures and direct Table A–5 saturation values plotted against absolute pressure with separate power-law fits.

Table 2: Power-law fit parameters for $T = AP^n$.

Dataset	A	n	R^2
Experimental measurements	31.218	0.2527	0.99904
Direct Table A–5 values	31.019	0.2557	0.99983

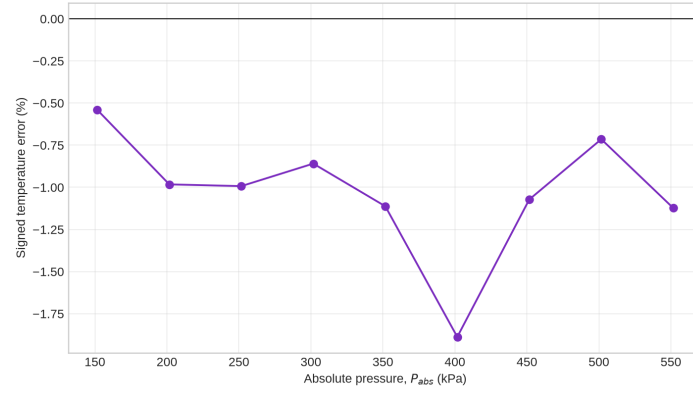


Figure 3: Signed experimental-temperature error relative to the point-specific interpolated saturation temperatures.

The mean signed error was -1.03% . Since every value was below the reference, the mean absolute error was also 1.03% . The maximum absolute error was 1.89% at 401.65 kPa absolute.

6 Discussion

6.1 Comparison and usefulness of the fitted equation

The two curves increased smoothly with absolute pressure and had very similar power-law parameters. The experimental exponent of 0.2527 differed from the direct-table value of 0.2557 by only 0.0030 . Both fits had $R^2 > 0.999$. Therefore, the apparatus captured the expected nonlinear relationship and its rate of increase over the measured range.

The experimental equation is useful as a compact empirical description of these measurements between 151.65 kPa and 551.65 kPa absolute. It should not replace steam tables in design work or be extrapolated outside the tested interval because it depends on the apparatus correction tables, measurement procedure and chosen fitting method.

6.2 Error trend

All signed errors were negative, indicating a small systematic tendency for the measured temperature to lie below the reference saturation temperature. The error did not increase monotonically with pressure: it ranged from -0.54% to -1.89% , peaked at 401.65 kPa, and then reduced. This pattern supports a small negative bias combined with point-to-point experimental variation, rather than a continuously growing pressure-dependent error.

6.3 Steam temperature, water temperature and assumptions

Saturated liquid and vapour have the same temperature only at thermodynamic equilibrium. The PT100 measured steam in the upper pipework rather than liquid inside the boiler. The comparison therefore made the following assumptions:

- air was sufficiently purged for total measured pressure to approximate the steam saturation pressure;
- pressure, boiler water, local steam and the PT100 had reached a common equilibrium value before each reading;
- atmospheric pressure remained effectively constant during the run;

- linear interpolation between adjacent resistance and steam-table entries was adequate over the small intervals used; and
- the Armfield bridge-correction table represented the console and sensor used in the experiment.

These assumptions are reasonable for processing the data, but the experiment did not test each one independently. Consequently, the possible causes below are interpretations consistent with the observed errors, not proven mechanisms.

Thermal lag occurs because energy must conduct through the fluid, pipework and probe sheath before the sensor reaches the local steam temperature. Heat loss from exposed pipework can also keep the probe region slightly cooler than the boiler. Either effect would produce negative temperature error if a reading were taken before complete stabilisation.

6.4 Possible sources of error and improvements

- **Residual air:** remaining non-condensable air contributes partial pressure, so the measured total pressure can exceed the steam partial pressure associated with the local saturation temperature.
- **Thermal lag and heat loss:** sensor response time and temperature gradients can make the upper-pipe measurement lower than the equilibrium boiler temperature.
- **Resistance conversion:** bridge correction, table resolution, rounding and linear interpolation influence every derived temperature.
- **Pressure measurement:** pressure-sensor calibration, display resolution and atmospheric-pressure uncertainty affect absolute pressure.
- **Single readings:** one value at each stage cannot distinguish random variation, hysteresis and repeatability from systematic bias.

The experiment would be stronger if each pressure stage were repeated after independent stabilisation. Means and standard deviations could then be reported, and an increasing-pressure run could be compared with a decreasing-pressure run to identify hysteresis. Longer air purging, an explicit stability criterion for both pressure and resistance, and documented instrument resolutions would further strengthen the uncertainty assessment.

7 Conclusion

The experiment reproduced the expected nonlinear increase in water saturation temperature with absolute pressure. After the original resistance readings were corrected and converted using the Armfield TH3 Data Sheets, the experimental fit was $T = 31.218P^{0.2527}$ with $R^2 = 0.99904$. This closely matched the direct steam-table fit, $T = 31.019P^{0.2557}$. Experimental temperatures were slightly below point-specific theoretical values at all nine stages, with a mean signed error of -1.03% and a maximum absolute error of 1.89% . The small negative bias is consistent with assumptions concerning air purging and equilibrium and with possible thermal lag, heat loss and instrument uncertainty. Repeated measurements are recommended to quantify repeatability and determine whether the remaining bias is systematic.

References

1. Y. A. Çengel, M. A. Boles, and M. Kanoğlu, *Thermodynamics: An Engineering Approach*, 9th SI ed., McGraw-Hill, 2020, Table A–5.
2. Armfield Ltd., *TH3 Saturation Pressure Apparatus Instruction Manual*, Issue 15, Data Sheets 1–2, pp. 22–24.
3. Auckland University of Technology, *ENME 601 Saturation Temperature & Pressure Lab Instructions*, 2025 update used for the 2026 laboratory.